

Performance Testing

1. Introduction

Performance testing is conducted to evaluate the responsiveness, stability, and efficiency of the Smart Lender – Loan Eligibility Prediction System under normal operating conditions. The objective is to ensure that the application provides accurate loan eligibility predictions with minimal response time while maintaining reliable performance.

2. Objectives

The performance testing aims to:

- Measure the application's response time.
 - Verify the stability of the Flask web application.
 - Ensure the machine learning model predicts results efficiently.
 - Validate the application's usability under normal workloads.
 - Identify any performance bottlenecks.
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3. Test Environment

Parameter	Details
Operating System	Windows 10/11
Programming Language	Python 3.x
Framework	Flask
Machine Learning Library	Scikit-learn
Development Tool	Visual Studio Code
Browser	Google Chrome
Version Control	Git & GitHub

4. Performance Test Scenarios

Test Scenario	Expected Result	Status
Application Launch	Home page loads successfully	Passed
Form Submission	User details are submitted without errors	Passed
Data Processing	Input data is validated and preprocessed	Passed
Model Prediction	Prediction generated successfully	Passed
Result Display	Loan approval/rejection displayed correctly	Passed

5. Performance Metrics

Metric	Result
Application Availability	100% during testing
Response Time	Less than 3 seconds
Prediction Accuracy	High (based on trained Random Forest model)
Error Handling	Successful validation of invalid inputs
User Experience	Smooth and responsive

6. Observations

- The web application loads successfully.
- User input is processed without delays.
- The Random Forest model generates predictions quickly.
- The application remains stable throughout testing.
- No significant performance issues were observed under normal usage.

7. Conclusion

The performance testing confirms that the Smart Lender system performs efficiently and reliably. The application responds quickly, processes applicant information accurately, and delivers loan eligibility predictions within an acceptable response time. The system is suitable for demonstration and further deployment.